

Deep Q-Learning-Based Handover for Spectral Coexistence Between Feeder and User Links in LEO Satellite Networks

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ABSTRACT Deploying feeder links (i.e., between satellites and ground stations) at Ka-band is becoming a popular option among satellite operators thanks to its consolidated technology maturity level. While using Ka-band for feeder links was not a major problem in geostationary satellites, its application in Low-Earth Orbit (LEO) and Medium-Earth Orbit (MEO) satellites brings new challenges due to the fast mobility of LEO/MEO platforms. In particular, the time-varying projection of the satellite spot-beam may result in overlapping with user link spot-beams and create inter-beam interference problems. Traditional beamforming strategies, while theoretically effective in mitigating interference by dynamically adjusting beam shapes, are often impractical due to the required recalculation speed and complexity overhead. In this work, we target a simpler solution based on a smart user-satellite assignment, i.e., handover strategy. A simple handover scheme offers an alternative but is limited by its reliance on accurate, instantaneous Carrier-to-Interference (C/I) values and its inability to adapt to dynamic environmental changes. To address these limitations, this paper proposes a novel handover scheme based on Deep Q-Network (DQN) reinforcement learning. The DQN-based approach enables adaptive learning and optimization of handover decisions based on real-time network conditions, thereby minimizing interference events and enhancing overall network performance. This method improves scalability, resource utilization, and robustness against prediction errors compared to traditional and conventional handover strategies. Comprehensive simulation results are presented, showcasing the proposed scheme's effectiveness in various scenarios and highlighting its advantages over existing methods. The findings suggest that the DQN-based approach offers a promising solution for managing feeder-user link interference in LEO/MEO satellite communications, ensuring robust and efficient connectivity.

INDEX TERMS Satellite communication, interference management, handover, deep Q-network (DQN), reinforcement learning.

I. INTRODUCTION

SATELLITES have long been considered key enablers for providing ubiquitous wireless coverage, addressing the growing demand for diverse applications and services either independently or as part of integrated

satellite-terrestrial networks [1], [2], [3]. Recently, Low-earth orbit (LEO) satellites have gained momentum with lower orbital altitudes, providing relatively low-latency communications, high network availability and increased flexibility when compared to the classical geostationary

satellites counterpart [4]. Some popular LEO constellations include the American SpaceX [5], the French Eutelsat-OneWeb [6], the European IRIS2 and the Chinese Qianfan (sometimes referred as G60) [7]. Additionally, the SES MEO O3b mPOWER constellation [8], which utilizes orchestrated handover and optional mid-band handover management to mitigate Gateway interference splash zones, is currently the only constellation in operation today in the Ka-band with such technology.

Operating a mobile mesh of small satellites at low-orbit requires a complex ground network composed of multiple gateways (teleports), ideally with sufficient geographically separation and regularly distributed around the coverage area such that there is high probability to have ground connection for each satellite on the fleet [9]. The links between satellite and ground gateways are called feeder links. Such feeder links have strong capacity requirements, as they are used to upload the aggregated data that then the satellites will redistribute to the users. Nowadays, satellite feeder links are based on RF communication, typically allocated in the Ka-band (26.5 – 40 GHz), which is also used by satellite-to-user links. Although some research is being devoted to optical communications for feeder links in the Q/V band [10], the technology is not yet mature enough and its practical deployment is limited. Ka-band feeder links are more convenient than Q/V band feeder links, as they face much less weather induced blockage events and, as a consequence, require less dense gateway network deployment.

In this work, we focus on the spectral coexistence between feeder and user downlink in a single-operator LEO satellite communication systems. Note that the inter-operator satellite interference is regulated by the International Telecommunication Union (ITU) under the Article 22 of the radio regulations and considered out of the scope of this work [11], [12], [13]. The Ka-band downlink portion available for fixed satellite services (FSS) falls within 17.7 GHz and 20.2 GHz, resulting in a 2.5 GHz bandwidth that is typically fully exploited by the gateway feeder downlink. The user equipment on the ground operates on the same spectrum portion but generally with much lower bandwidth, i.e., ~ 200 MHz. For the sake of clarity, the spectral representation is sketched in Fig. 1. It becomes evident that the impact of the user interference on the feeder link (gateway receiver) will be minimal due to the differences on the bandwidth of such two links. On the other hand, the user link might suffer the effects of strong interference as the whole user link bandwidth will be impacted.

The potential interference scenarios under consideration of this work are schematically depicted in Fig. 2. In particular, the feeder link and the user link can be emitted from different satellite (Fig. 2(a)) or from the same satellite using different spot-beams (Fig. 2(b)). Obviously, the second case poses more challenges given that the angular separation between the interference and the desired signal is minimal. The latter is a situation to be avoided. The simple example included

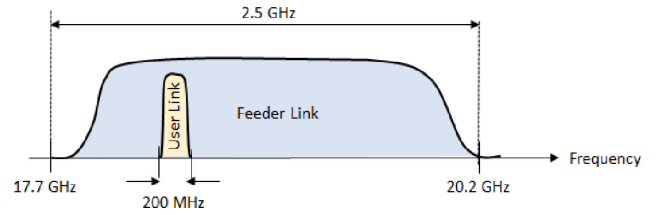


FIGURE 1. Spectrum representation of the feeder and user link coexistence.

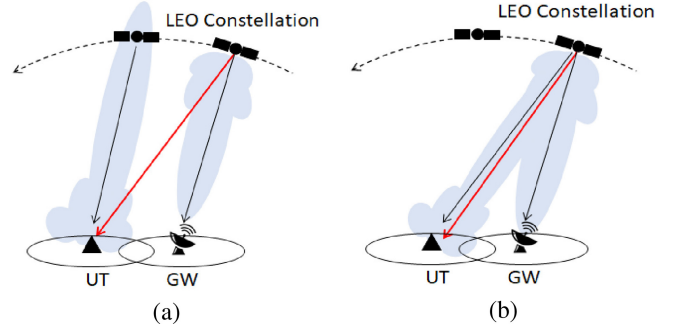


FIGURE 2. Spectrum coexistence between feeder and user link: (a) Connection via different satellite, (b) Connection with the same satellite.

in Fig. 2 evidenced that a proper satellite selection might avoid strong interference events.

The spectrum sharing between feeder links and user links can lead to interference issues due to the spatial proximity of these links, significantly degrading the quality of service (QoS) and constraining overall network capacity.

A. LITERATURE REVIEW AND CONTRIBUTIONS

The topic of feeder link and user link sharing has not been widely investigated but the problem has certain similarities with frequency reuse for user link (i.e., between users). Therefore, the first step is to revise and clarify the conventional techniques that have been applied for simultaneous multi-user service. The obvious solution is to keep the two on-ground receivers that share the same frequency sufficiently separated. The latter is commonly known in the community as frequency-reuse schemes [14] and it is illustrated in Fig. 3(a). While this can be a valid option for user links that can satisfy the demand with partial available user bandwidth, the data-hungry feeder downlink claims for full-frequency operation. Therefore, the frequency-reuse schemes do not apply to the considered scenario.

On the other hand, some work like [15] have introduced the spatial isolation concept, which for instance in [15] was considered to separate 5G terrestrial cells from 5G satellite terminals. The spatial separation is illustrated in Fig. 3(b). Geographically separating feeder and user links can be a valid option but it is in general not possible when dealing with gateway ground stations. In particular, gateway locations are generally surrounded by multiple user terminals as they are strategically allocated within the satellite coverage area. Satellite operators' customers require to have the gateway ground stations closer to them for security and

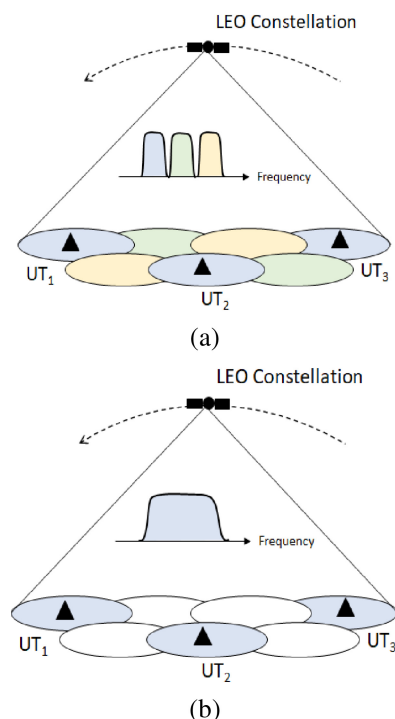


FIGURE 3. Methods to deal with user link interference: (a) Frequency reuse ensuring enough physical distance between beams, (b) Spatial isolation ensuring that active beams are geographically separated.

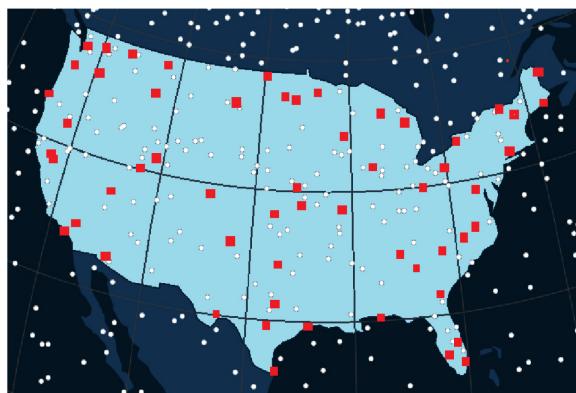


FIGURE 4. Starlink gateway deployment in United States of America (red squares). White points indicate orbiting satellite. Information extracted from <https://satellitemap.space/>.

reliability reasons (e.g., in the same country or in friendly territory). Therefore, it is assumed that spatial separation is in general not a viable option to deal with the feeder and user link interference. Take for example Figure 4, which shows the tentative gateway locations of Starlink over the United States of America (USA) (information extracted from <https://satellitemap.space/>, which is based on FCC filings and other sources). From Fig. 4 it is evident that the geographical separation of users and gateways is challenging.

Another popular technique for interference management in satellite systems is antenna beamforming [16], [17], [18]. Traditional beamforming strategies aim to mitigate the

interference by dynamically adjusting the beam shape, focusing the satellite's transmission power in desired directions to minimize interference with neighboring beams. While effective in theory, these methods can be impractical due to several significant limitations. The primary challenge lies in the computational overhead required to continuously recalibrate the beamforming parameters in real-time. This involves complex algorithms and high processing power, which can be resource-intensive and lead to increased operational costs [19]. Additionally, the dynamic nature of LEO satellite networks, with their rapidly changing topology and user distribution, exacerbates these computational demands, making real-time beamforming adjustments difficult to implement efficiently [1], [20].

In contrast, a simple satellite handover scheme can address the problem without adding complexity in the satellite beamforming network. In our previous work [21], we presented a satellite handover strategy offering an alternative solution to the feeder-user link spectrum sharing by dynamically switching connections based on predefined thresholds. This scheme was designed to alleviate interference by transitioning a UT's communication link from one satellite to another when certain conditions are met. While this approach showed promising results, it has several limitations. The scheme depends heavily on the availability of accurate and instantaneous C/I values to compare with the specified threshold. In practical scenarios, accurately predicting C/I values can be challenging due to the dynamic nature of the satellite environment and the variability in interference levels. The simple handover scheme operates based on predefined thresholds and does not adapt to changes in the environment. This static nature means it cannot respond to unforeseen variations in network conditions, such as sudden changes in user density and interference patterns. As a result, the scheme may either fail to initiate a necessary handover in time or trigger unnecessary handovers, both of which can degrade overall network performance.

Several simple user-link handover strategies have been proposed in the literature that disregard the potential interference between gateways and user terminals. These methods often prioritize other factors, such as maintaining connectivity or minimizing handover frequency, without fully accounting for the complexities of interference. Examples include the maximum service time or visibility-based handover [22], which selects the satellite offering the longest remaining visibility, and the minimum distance-based handover [23], which chooses the satellite closest to the user terminal. While these approaches can be effective in certain scenarios, they often fall short in environments where interference management is critical, potentially leading to suboptimal network performance and reduced quality of service for users.

Recent advancements in machine learning, particularly in reinforcement learning (RL), have opened new avenues for addressing these challenges. RL, with its ability to learn optimal policies through interaction with dynamic

environments, has been applied to various decision-making problems in satellite communications, including resource allocation and load balancing [24], [25]. In particular, deep Q-network (DQN), a prominent RL technique, has shown significant potential to tackle the generic handover problem in satellite networks. For example, the authors in [26], [27] propose DRL-based algorithms to improve handover decisions in general non-terrestrial network (NTN) settings. However, these studies primarily focus on maximizing connectivity or balancing load without explicitly addressing interference-related challenges. In contrast, our work targets a unique and increasingly relevant problem: managing spectral coexistence and interference between user terminals (UTs) and gateway stations (GWs) that operate in overlapping frequency bands in LEO constellations. Our DQN-based handover model is specifically designed to minimize the impact of co-channel interference caused by the proximity of GWs and UTs, an aspect not considered in the aforementioned works. By integrating interference-aware metrics into the state and reward structure, our solution directly addresses the link degradation challenges that arise in dense LEO deployments, offering a practical and scalable approach for real-time interference mitigation.

In this work, we propose a DQN-based handover scheme to address these limitations and further improve handover performance. By employing DQN, the proposed handover scheme can adaptively learn and optimize handover decisions based on real-time network conditions, thereby minimizing interference and enhancing overall network performance. This adaptive approach allows the handover mechanism to respond dynamically to changes in the environment, improving scalability, resource utilization, and robustness against prediction errors. For instance, in the illustrative scenario of Figure 5, the user terminal could switch from satellite 2 to satellite 3 to reduce the interference.

The main contributions of this paper are as follows:

- We present and address a novel interference scenario that is becoming more and more relevant to the LEO satellite constellations, which is the spectrum sharing between bandwidth-hungry gateway links and user links.
- We characterize the interference scenario and propose a novel handover mechanism leveraging deep Q-network reinforcement learning to manage the gateway-user link interference in LEO satellite communication systems. The interference dynamics are explicitly modeled via the C/I metric and directly influence the learning process, making our solution tailored for this unique scenario. The algorithm's reward structure promotes both high link quality and handover stability, thus addressing the core challenge of managing handovers in the presence of strong interlink interference. This approach allows for adaptive decision-making based on real-time network conditions. Unlike handover methods that require predicting the C/I values before making a

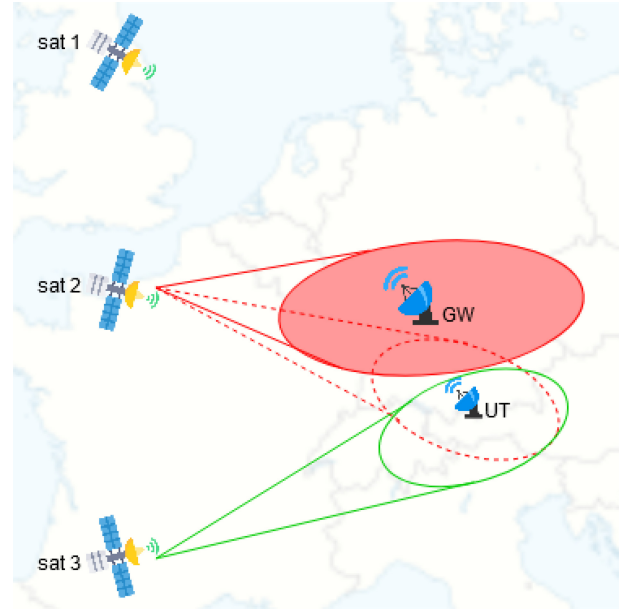


FIGURE 5. Illustration of the interference problem.

handover decision, the proposed DQN-based approach calculates the C/I only after the handover decision is executed. This allows the system to learn optimal handover policies based on observed outcomes rather than relying on potentially inaccurate C/I predictions. As a result, the adaptive nature of the DQN algorithm makes the system more resilient to environmental variations and prediction inaccuracies.

- We provide extensive simulation results demonstrating the effectiveness of the proposed scheme under various scenarios, highlighting its advantages over traditional handover methods.

The remainder of this paper is organized as follows. Section II details the system model and problem formulation. Section III describes the proposed DQN-based handover scheme. Section IV presents the simulation results and performance analysis. Finally, Section V concludes the paper and outlines future research directions.

II. SYSTEM MODEL

We consider a downlink communication scenario involving N_S LEO satellites serving N_{GW} GWs and N_{UT} UTs. The system model is based on a Walker-Star constellation, although the proposed approach can be generalized to other satellite constellation architectures. Each LEO satellite moves in a predictable orbit, allowing the UTs to know the set of satellites available for communication and its remaining visibility time [28]. The remaining visibility time T_{vis} for a satellite is determined by:

$$T_{vis} = T_{max} - T_{elapsed}, \quad (1)$$

where T_{max} is the total predicted visibility duration of a given satellite to the UT, measured from the moment the



FIGURE 6. Example of satellite visibility timeline.

satellite first becomes visible until the time it exits the UT's visibility, and T_{elapsed} is the elapsed time since the satellite first became visible. A visual example of this timeline is shown in Figure 6, where the satellite becomes visible to the UT at time t_0 and remains in view until time $t_0 + T_{\text{max}}$. At any intermediate time t , the elapsed time is $T_{\text{elapsed}} = t - t_0$, and the remaining visibility time is given by equation (1).

At any given time t , multiple satellites may simultaneously cover a specific user k . Coverage is defined as the satellite being within the user's visibility region, which is determined by a minimum elevation angle $\theta_{\min}$. A satellite m_k^t is considered to cover user k if the elevation angle $\theta_{s,k}$ between the satellite and the user satisfies $\theta_{s,k} \geq \theta_{\min}$. The set of satellites providing coverage to user k at time t can be represented as:

$$\mathcal{M}_k^t = \{m_k^t \in \{1, 2, \dots, N_S\} \mid \theta_{s,k}(t) \geq \theta_{\min}\}, \quad (2)$$

where N_S is the total number of satellites in the constellation. This set includes all satellites that meet the minimum elevation angle requirement and thus can establish a communication link with user k at time t . However, we assume that each UT will always be connected to a single satellite at each instant t .

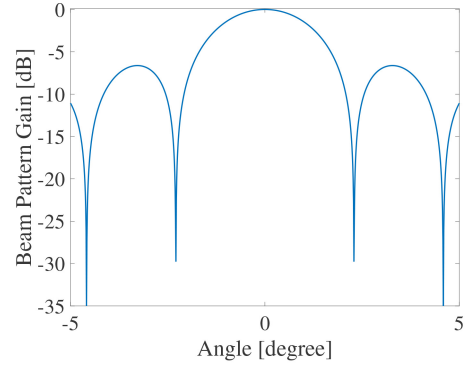
The predictable motion of LEO satellites enables the system to anticipate the set of satellites providing coverage for each user in advance. By dynamically selecting the optimal satellite from the set $\mathcal{M}_k^t$, the system aims to minimize received interference and enhance received signal quality, thereby improving the overall carrier-to-interference-and-noise ratio (CINR) and ensuring better network performance.

A. SATELLITE AND GROUND STATION CHARACTERISTICS

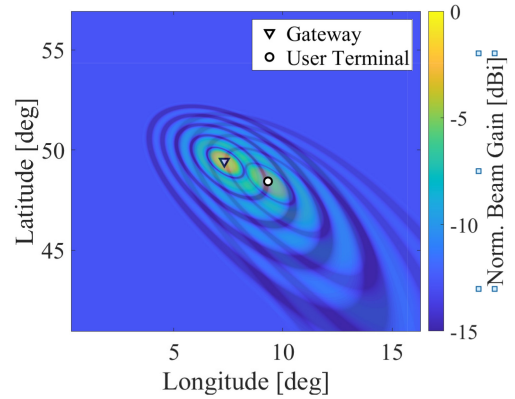
The GWs are equipped with high-gain antennas capable of operating across the entire bandwidth range of 17.7 GHz to 20.2 GHz in the Ka-band.¹ This wide bandwidth is typical for feeder links, which aggregate traffic from multiple UTs and facilitate efficient data transmission to and from the satellite network. The UTs operate within the same frequency spectrum but with a narrower bandwidth, denoted as B_{UT} , where $B_{\text{UT}} < B_{\text{GW}}$. Here, B_{GW} refers to the total bandwidth allocated for feeder link operations. This reduced bandwidth allocation reflects the lower individual data throughput requirements of UTs compared to the aggregate traffic handled by GWs.

The satellites utilize pencil-shaped beams to minimize interference, with each beam designed to cover specific

¹Although there are prospects of exploiting Q/V band, the technology is not yet mature enough and requires a very dense ground network deployment to counteract weather impairments, thus not considered in this work.



(a) Satellite beam pattern in the angular domain.



(b) Satellite beam pattern projected on the Earth's surface.

FIGURE 7. (a) Satellite beam pattern generated with a planar array and (b) Projected beams at GW and UT.

ground stations while reducing overlap with neighboring beams. The beam generation uses a Direct Radiating Array (DRA) with a planar array configuration, as described in [19]. The resulting antenna beam pattern is shown in Figure 7(a). Figure 7(b) shows the projection of the DRA-generated beam pattern when centered over the GW and UT locations, where the projected beam at GW interferes with the projected beam at UT. As discussed in the introduction, the narrow-beam design should prevent interference between beams. However, practical scenarios can deviate from this ideal situation, particularly when UTs are in close proximity to GWs. In such cases, the pencil-type beams may “splash” due to the projection over the Earth curvature, leading to unintended overlap and potential interference.

B. CHANNEL MODEL

The channel model at high frequency bands (i.e., Ka band) mainly includes the path loss and off-axis angle antenna loss [29]. The free-space path loss due to the distance between the satellite and ground stations is modeled using the free-space path loss formula:

$$L_{\text{FS}}[\text{dB}] = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}\left(\frac{4\pi}{c}\right), \quad (3)$$

where d is the distance between the satellite and the ground station, f is the frequency, and c is the speed of light.

The off-axis gain loss represents the reduction in antenna gain when the signal is not directed at the boresight of the antenna. It is important to distinguish between two types of channels in our system model: the intended communication channel and the interference channel. The intended channel assumes that the UTs and GWs are always positioned at the beam center, ensuring maximum gain for their respective links. However, interference between GW and UT links is modeled by considering the off-axis angle between them, which affects the interference power received. The off-axis gain loss can be quantified as:

$$L_R[\text{dB}] = 12 \left(\frac{\theta_R}{\theta_{3\text{dB}}} \right)^2, \quad (4)$$

where θ_R is the off-axis angle and $\theta_{3\text{dB}}$ is the half power beamwidth of the receive radiation pattern [30].

C. PERFORMANCE METRIC

The quality of the communication link is assessed using the Carrier-to-Interference power Ratio (C/I), defined as:

$$C/I[\text{dB}] = 10 \log_{10} \left(\frac{P_{\text{UT}}}{P_{\text{TOT}}} \right), \quad (5)$$

where P_{UT} represents the intended received signal power at the UT, and P_{TOT} denotes the total interference signal power in Watts. The intended received signal power P_{UT} in dBW is given by:

$$P_{\text{UT}}[\text{dBW}] = \text{EIRP}_{\text{UT}}[\text{dBW}] - L_{\text{UT}}[\text{dB}] + G_{\text{rx, UT}}[\text{dBi}], \quad (6)$$

where EIRP_{UT} is the Effective Isotropic Radiated Power for the UT, L_{UT} is the path loss to the UT, and $G_{\text{rx, UT}}$ is the composite receive gain at the UT. The composite receive gain G_{rx} is expressed as:

$$G_{\text{rx}}[\text{dBi}] = G_{\text{Rmax}}[\text{dBi}] - L_R[\text{dBi}], \quad (7)$$

where G_{Rmax} is the antenna receive gain at boresight, and L_R is the off-axis gain loss in equation (4).

The interference signal power S_I from a ground station is given by:

$$S_I[\text{dBW}] = \text{EIRP}_I[\text{dBW}] - L_I[\text{dB}] + G_{\text{rx, I}}[\text{dBi}] + 10 \log_{10} \left(\min \left(1, \frac{B_{\text{UT}}}{B_I} \right) \right), \quad (8)$$

where EIRP_I is the EIRP for the interfering carrier, L_I is the path loss, $G_{\text{rx, I}}$ is the composite receive gain for the interfering carrier, B_{UT} is the carrier bandwidth for the UT, and B_I is the interfering carrier bandwidth.

D. BENCHMARK HANDOVER MECHANISMS

The handover process is designed to ensure seamless communication as UTs move between coverage areas of different satellites. In the context of this work, we consider two types of benchmark handover criteria: (i) Carrier-to-Interference

Ratio (C/I)-based handover, and (ii) Maximum remaining visibility-based handover. Another common handover criterion is the minimum distance-based approach, where the user terminal always connects to the closest satellite. However, as studied in [21], this method is highly sensitive to pointing errors, making it less reliable for practical deployment in LEO satellite networks. For this reason, we focus on C/I and visibility-based handovers as more robust benchmark strategies.

1) C/I-BASED HANDOVER

The C/I-based handover criterion selects the satellite that provides the highest Carrier-to-Interference Ratio (C/I) to the UT. This criterion aims to optimize the signal quality by minimizing the interference experienced by the UT. The C/I is defined in equation (5).

In this traditional handover strategy [21], the C/I is compared to a predetermined threshold $(C/I)_{\text{thr}}$, where:

$$\begin{aligned} \text{Link is at risk: } & (C/I) < (C/I)_{\text{thr}}, \\ \text{Link has no risk: } & (C/I) \geq (C/I)_{\text{thr}}, \end{aligned} \quad (9)$$

The satellite providing the highest C/I is selected for handover, ensuring that the UT is always connected to the satellite offering the best signal quality and least interference. This approach is particularly effective in scenarios with significant interference from other sources. However, this method requires an estimation of C/I before making the handover decision, which introduces several challenges:

- *Difficulty in predicting C/I:* The C/I value at the new satellite must be estimated before the handover occurs. However, interference power from co-channel beams cannot be directly measured in advance and must be inferred from pilot signals, making the estimation process complex and prone to errors.
- *Estimation inaccuracies:* Even when interference can be estimated, inaccuracies in the prediction can lead to suboptimal handover decisions, especially in highly dynamic environments where interference fluctuates rapidly.
- *High monitoring overhead:* This approach requires continuous real-time monitoring of C/I across multiple satellites, introducing significant computational and signaling overhead.
- *Reactive decision-making:* The method only initiates handover after the C/I threshold has already been crossed, meaning the link is already at risk before corrective action is taken. This reactive nature can lead to service degradation, unnecessary handovers, or communication disruptions.

By contrast, the reinforcement learning (RL)-based approach does not require predicting C/I in advance but instead learns optimal handover policies by evaluating the actual C/I observed after handovers, allowing for more adaptive and proactive decision-making.

2) MAXIMUM REMAINING VISIBILITY-BASED HANDOVER

The maximum remaining visibility-based handover criterion selects the satellite that will remain visible to the UT for the longest duration. This criterion focuses on reducing the frequency of handovers by prioritizing the satellite with the maximum remaining visibility time. The visibility time is calculated based on the satellite's orbital parameters (which are periodically broadcasted by the satellite system) and the UT's location, which may be determined via GNSS or other location estimation methods. By selecting the satellite with the longest remaining visibility time, the system minimizes the handover occurrences, thereby reducing the potential for service interruptions and signaling overhead. While a properly executed handover should not result in noticeable disruptions, any degradation in service may still occur, depending on factors such as handover frequency and the capabilities of the base station.

The remaining visibility time T_{vis} for a satellite is obtained as in equation (1). The satellite with the highest T_{vis} is selected for handover, ensuring prolonged connectivity and stability in the communication link.

Both handover criteria have their own advantages and are suitable for different scenarios. The C/I-based handover is beneficial in environments with high interference, as it ensures the best possible signal quality. On the other hand, the Maximum Remaining Visibility-based handover is advantageous in scenarios where reducing the number of handovers is critical, thereby enhancing the stability and reliability of the communication link.

In practice, a hybrid approach that considers both C/I and remaining visibility could be implemented to balance signal quality and handover frequency, providing an optimal solution for dynamic and heterogeneous satellite communication environments.

III. PROPOSED DQN-BASED HANDOVER

A. INTRODUCTION TO DEEP Q-NETWORK

RL is a machine learning algorithm where an AI agent learns optimal behaviors through interactions with its environment. The agent takes actions, observes the resulting state and receives feedback in the form of rewards from the environment. The primary objective of the agent is to maximize its cumulative reward over time.

In Q-learning, a popular RL algorithm, the Q-value, denoted as $Q_t(s, a)$, quantifies the expected total reward of choosing action a in state s at time t . The essence of Q-learning lies in learning the optimal action-value function, which helps the agent make the best possible decision at each state. As training progresses, the algorithm iteratively updates the Q-values for all state-action pairs, storing them in a data structure known as the Q-table. The Q-learning update rule is given by:

$$Q_{t+1}(s, a) = (1 - \alpha)Q_t(s, a) + \alpha \left(r + \gamma \max_{a'} Q_t(s', a') \right), \quad (10)$$

where s' is the state reached after taking action a from state s , r is the immediate reward obtained after taking action a in state s , α is the learning rate, and γ is the discount factor. The term $\max_{a'} Q_t(s', a')$ represents the maximum Q-value for the next state s' , considering all possible actions a' . This ensures the agent is always looking ahead to maximize future rewards. Over numerous iterations, the Q-table evolves to accurately reflect the optimal policy, enabling the agent to select the best action for any given state.

However, in environments with large state-action spaces, the Q-table can grow exponentially, quickly becoming impractical to store and manage. This is where Deep Q-Networks (DQNs) come into play [31].

A DQN integrates Q-learning with deep neural networks to approximate the Q-value function, thus eliminating the need for a large Q-table. Instead of explicitly storing the Q-values for every state-action pair, a neural network is trained to predict Q-values. The network takes the current state as input and outputs Q-values for all possible actions. The action corresponding to the highest Q-value is then selected. This method allows for efficient handling of high-dimensional state spaces with discrete action sets, making RL feasible in complex environments.

B. PROBLEM FORMULATION

At each time step t , a user terminal k must select a serving satellite m_k^t from its set of visible satellites $\mathcal{M}_k^t$. The objective is to select a sequence of serving satellites $\{m\}_{t=1}^T$ that maximizes the overall link quality while minimizing unnecessary handovers. The optimization problem can be expressed as:

$$\begin{aligned} \max_{\{m_k^t\}} \quad & \sum_{t=1}^T \left(\omega_1 \cdot C/I[\text{dB}]_t^k - \omega_2 \cdot \delta(m_k^t \neq m_k^{t-1}) \right) \\ \text{s.t.} \quad & m_k^t \in \mathcal{M}_k^t, \quad \forall t, \\ & \theta_{m_k^t, k}(t) \geq \theta_{\min}, \quad \forall t, \end{aligned} \quad (11)$$

where:

- $C/I[\text{dB}]$ is defined in equation (5);
- $\delta(m_k^t \neq m_k^{t-1})$ is an indicator function equal to 1 if a handover occurs at time t ;
- ω_1 and ω_2 are weighting factors balancing link quality and handover cost;
- $\mathcal{M}_k^t$ is the set of satellites covering user k at time t ;
- $\theta_{m_k^t, k}(t) \geq \theta_{\min}$ ensures that the selected satellite has a sufficient elevation angle for maintaining a viable communication link.

This optimization problem is non-trivial due to the dynamic and time-varying nature of satellite visibility, link quality, and interference. It naturally fits into a Markov Decision Process (MDP) framework, which motivates the use of Deep Q-Learning. The DQL agent learns a policy that approximates the optimal satellite selection strategy by interacting with the environment, observing transitions,

and optimizing long-term returns as expressed in the above objective.

To solve this problem, we use a DQN agent to learn an optimal handover policy over time based on observed state transitions and reward feedback. In our DQN framework, the state space is designed to capture the key parameters that influence handover decisions. The state s_t at time t is defined by the remaining visibility times of the V closest visible LEO satellites. Specifically, for each of these V satellites, the remaining visibility time is denoted as $T_{i,t}$, where i represents the index of the satellite. The state vector s_t is then computed as:

$$s_t = [T_{1,t}, T_{2,t}, \dots, T_{N,t}]. \quad (12)$$

By incorporating the visibility times, the Q-learning model learns the long-term impact of selecting satellites with longer visibility, optimizing handover decisions by balancing C/I maximization with handover frequency reduction for a more adaptive strategy. This choice reflects information that is locally available and predictable in LEO constellations, enabling distributed and low-latency decision-making. However, we acknowledge that this simplified state representation does not capture other potentially relevant factors, such as real-time interference measurements, gateway traffic load, UT mobility, or the spatial distribution of interference sources. While integrating such features into the state space could enhance the policy's ability to react to more complex dynamics, doing so would require reliable estimation mechanisms, additional signaling, or increased computational overhead, which is especially challenging in real-time or decentralized settings.

The action space A consists of all possible handover decisions that the DQN agent can make. These actions include:

- A_{up} : This action initiates a handover to a neighboring LEO satellite positioned above the current satellite in the sky. It is typically chosen when the current satellite's visibility is about to end, or its link quality is degrading.
- A_{down} : This action involves switching to a neighboring LEO satellite positioned below the current satellite. This decision may be made when the satellite above is not optimal, or the satellite below offers better communication conditions.
- A_{stay} : This action maintains the existing communication link with the current serving satellite. It is chosen when the current satellite continues to offer acceptable link quality and its visibility is still sufficient.

This approach follows a distributed design, where each UT independently determines when and to which satellite to hand over based solely on its local measurements and learned policy. While multiple UTs are present in the simulated environment, the proposed DQN-based strategy is applicable to any UT operating independently, without requiring coordination with others.

A centralized coordination mechanism among UTs is not assumed in this work, as it is fundamentally constrained

by the architectural and latency limitations of non-terrestrial networks. Specifically, centralized control would require round-trip signaling between UTs and a common control entity, such as a satellite or satellite-plus-gateway combination, which introduces significant propagation delays and signaling overhead. These delays could compromise the timeliness of the handover actions, especially in dynamic LEO environments where interference and topology evolve rapidly.

Therefore, rather than being a design choice for simplicity, the distributed nature of our approach is motivated by the practical infeasibility of coordinated decision-making under such constraints.

To compute the reward r in equation (10), we apply the reward function $R(s_t, a_t)$, which provides immediate feedback on the quality of handover decisions in the satellite communication system. The reward reflects the short-term outcome of the decision. In contrast, the Q-value $Q_t(s, a)$ represents the expected long-term reward from selecting action a in state s , considering both immediate and future rewards.

The reward function has two key components: the reward for improvement in the C/I and the stability reward, which accounts for changes in the serving satellite. The overall reward function can be expressed as:

$$R(s_t, a_t) = R_{C/I} + R_{\text{stability}}, \quad (13)$$

where $R_{C/I}$ represents the reward associated with the enhancement in the C/I ratio and $R_{\text{stability}}$ reflects the stability of the link, considering whether the serving satellite has changed compared to the previous time step.

The reward for improving the C/I ratio is given by:

$$R_{C/I} = \frac{C/I_{\text{post-HO}} - C/I_{\text{min}}}{C/I_{\text{min}}}, \quad (14)$$

where $C/I_{\text{post-HO}}$ represents the Carrier-to-Interference ratio of the newly selected satellite at the current time step, measured after the handover decision has been executed, and C/I_{min} is the minimum threshold C/I ratio, serving as a baseline for acceptable performance. It is a normalized value that reflects how much the current C/I ratio improves relative to the threshold.

The stability reward

$$R_{\text{stability}} = \begin{cases} -1 & \text{if } C/I_{\text{post-HO}} > C/I_{\text{min}} \text{ and different sat} \\ +1 & \text{if } C/I_{\text{post-HO}} > C/I_{\text{min}} \text{ and same sat} \\ 0 & \text{otherwise} \end{cases} \quad (15)$$

adjusts the reward based on whether the satellite serving the UT has changed. A penalty of -1 is assigned if the C/I ratio is above the threshold but the serving satellite has changed from the previous time step. This discourages frequent satellite switching which can disrupt the connection. A reward of $+1$ is given if the C/I ratio is above the threshold and the same satellite continues to serve the UT. This promotes stability and continuity in the link. If the C/I

ratio does not exceed the threshold, no additional reward or penalty is applied.

The proposed DQN-based handover strategy is designed to minimize real-time signaling overhead during deployment. While the C/I ratio is used during training to compute the reward and guide learning, it is not required as a continuous input during operation. The DQN agent bases its decisions on satellite visibility information, which can be derived from orbital predictions and does not require constant measurement reporting or feedback. This design contrasts with conventional C/I-threshold-based handovers, which require frequent monitoring and signaling whenever C/I degrades. Our approach avoids this control-plane burden by embedding the learned decision policy into the terminal. Once training is complete, handover decisions are made locally at the UT using only the current state, thus enabling fast and lightweight inference suitable for dynamic LEO constellations.

C. DQN TRAINING PROCESS

The training process for a DQN involves several sequential steps designed to enable the agent to learn optimal decision-making strategies. Initially, the algorithm starts by setting up a replay memory to store experiences, where each experience consists of a state, action, reward, and the resulting next state. A single neural network, referred to as the Q-network, is then initialized. This network will both predict Q-values and be used to compute target values during training. The Q-network architecture is composed of one input layer, one hidden layer, and one output layer. The input layer receives a V -dimensional state vector representing the remaining visibility durations for the V closest visible satellites. This is followed by a fully connected hidden layer with 24 neurons using the ReLU activation function. The output layer is a dense layer with three neurons, each corresponding to one of the possible discrete actions: handover upward, handover downward, or stay. The network is trained using the Adam optimizer with a learning rate of 0.9, and the mean squared error (MSE) is used as the loss function.

During each training step, the agent interacts with the environment by selecting actions based on an epsilon-greedy policy. This policy involves choosing a random action with probability ϵ to encourage exploration and selecting the action with the highest predicted Q-value from the Q-network with probability $1 - \epsilon$ to encourage exploitation. After performing the action, the agent observes the immediate reward and the resulting new state. Figure 8 illustrates the ϵ -greedy decision-making process used to balance exploration and exploitation during training.

The experience, represented as a tuple (s, a, r, s') , is stored in the replay memory. To update the Q-values, the algorithm samples a mini-batch of experiences from the replay memory. For each experience in the mini-batch, the target value y_i is computed. If the next state s' is terminal, the target value is simply the immediate reward r_i . If the next state s' is non-terminal, the target value is computed as the immediate

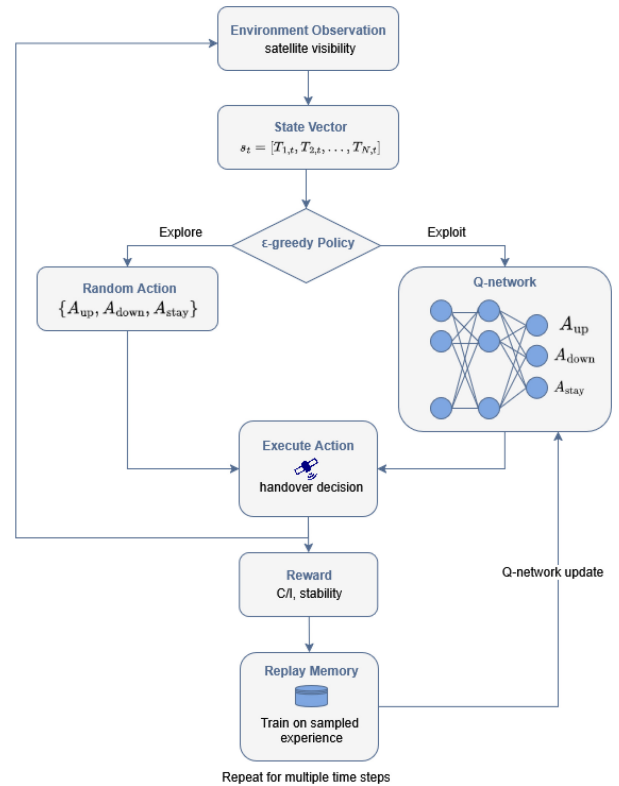


FIGURE 8. Illustration of ϵ -greedy decision-making policy used in the DQN training process.

reward r_i plus the discounted maximum Q-value of the next state, where the Q-values are predicted by the same Q-network. Specifically, the target value y_i is given by:

$$y_i = \begin{cases} r_i & \text{if } s' \text{ is a terminal state} \\ r_i + \gamma \max_{a'} Q(s', a') & \text{if } s' \text{ is not a terminal state} \end{cases} \quad (16)$$

where γ is the discount factor and $Q(s', a')$ denotes the Q-values predicted by the Q-network for the next state s' .

The loss function is then computed as the mean squared error between the target values y_i and the Q-values predicted by the network for the taken actions. The loss L is given by:

$$L = \frac{1}{N} \sum_{j=1}^N (y_j - Q(s_j, a_j))^2, \quad (17)$$

where N is the mini-batch size and y is defined in equation (16). Gradient descent is used to minimize this loss, thereby updating the Q-network's weights to better approximate the optimal Q-values.

The process continues with the network being updated and the exploration rate ϵ being decayed over time to shift from exploration towards more exploitation of the learned policy. This iterative training continues until the agent's performance stabilizes, indicating that the Q-network has effectively learned to predict Q-values that reflect the optimal policy for decision-making.

Algorithm 1 DQN-Based Handover Strategy

```

1: Initialize Q-network with 1 hidden layer (24 ReLU
   units), output size = number of actions
2: Set hyperparameters:  $\gamma = 0.95$ ,  $\epsilon = 1.0$ ,  $\epsilon_{\min} = 0.01$ ,
    $\epsilon_{\text{decay}} = 0.99$ 
3: Initialize memory buffer with capacity = 2000
4: for each time step  $t$  do
5:   Observe current state  $s_t$ 
6:   With probability  $\epsilon$ , select random action  $a_t$ , else
      $a_t = \arg \max_a Q(s_t, a)$ 
7:   Execute  $a_t$ , observe next state  $s_{t+1}$  and reward  $r_t$ 
8:   Store transition  $(s_t, a_t, r_t, s_{t+1})$  in memory
9:   if memory has enough samples then
10:    Sample minibatch of transitions
11:    for each transition  $(s, a, r, s')$  in batch do
12:       $y = r + \gamma \cdot \max_{a'} Q(s', a')$ 
13:      Update Q-network by minimizing  $(y - Q(s, a))^2$ 
14:    end for
15:  end if
16:  Update  $\epsilon \leftarrow \max(\epsilon \cdot \epsilon_{\text{decay}}, \epsilon_{\min})$ 
17: end for

```

TABLE 1. DQN training parameters.

Parameter	Value
State vector dimension ($ \mathcal{S} $)	10
Number of actions ($ \mathcal{A} $)	3 (stay, up, down)
Hidden layer size	24 neurons
Activation function	ReLU
Output layer	Linear, 3 neurons
Optimizer	Adam
Learning rate (α)	0.1
Discount factor (γ)	0.95
Initial exploration rate (ϵ)	1.0
Minimum exploration rate ($\epsilon_{\min}$)	0.01
Exploration decay rate	0.99
Batch size	4
Loss function	Mean Squared Error (MSE)

Algorithm 1 outlines the key steps of the proposed Deep Q-Network training procedure, including state selection, action generation using an ϵ -greedy policy, experience replay, and Q-value updates. In parallel, Table 1 summarizes the main hyperparameters used in the training process, including learning rate, discount factor, replay buffer size, and neural network configuration.

The proposed DQN-based handover strategy includes several adaptations tailored to the LEO satellite context. First, the state representation captures the remaining visibility times of the V closest satellites, allowing the agent to make proactive decisions that anticipate satellite departure. Second, the reward function jointly considers communication quality (via a normalized C/I improvement) and link stability (via penalties for unnecessary handovers), balancing performance and continuity. Lastly, the action space is structured into

three intuitive decisions—stay, handover up, and handover down—which reflect practical handover directions in a LEO constellation.

IV. SIMULATION RESULTS

To assess the performance of the proposed DQN-based handover scheme, we conducted simulations within a realistic environment that replicates the dynamics of LEO satellite communications. For comparison, we also evaluated the two baseline handover strategies presented in Section II-D. In particular, the C/I-based handover strategy selects the satellite that offers the highest signal quality relative to interference, ensuring the best possible link quality at each moment. The Best C/I-based strategy assumes ideal conditions where the C/I for all candidate satellites is known in advance before the handover decision. This makes it an unrealistic upper-bound in practice, but it serves as a valuable reference for benchmarking the proposed DQN-based method, which achieves comparable performance without requiring oracle-level knowledge. On the other hand, the visibility-based handover approach prioritizes the satellite with the longest remaining visibility time, aiming to minimize the frequency of handovers by maintaining connectivity with a satellite for as long as possible.

The simulation scenario is modeled using a Walker Delta-Star constellation, comprising 12 orbital planes, each containing 20 LEO satellites positioned at an altitude of $h_{\text{sat}} = 600$ kilometers. The system operates at a frequency of $f_{\text{sat}} = 19.5$ GHz. The satellites are equipped with antennas employing a planar array configuration of 50×50 elements, providing an Effective Isotropic Radiated Power Spectral Density of $EIRP_{\text{sat}} = -48$ dBW/Hz. Figure 7(a) illustrates the obtained beam pattern. The GW link operates with a bandwidth of $B_{\text{GW}} = 2$ GHz, while the UT link is allocated a bandwidth of $B_{\text{GW}} = 200$ MHz. The GWs and UTs are equipped with antenna dishes of $D_{\text{GW}} = 3.5$ meters and $D_{\text{UT}} = 0.6$ meters in diameter, respectively. A minimum elevation angle of $\theta_{\min} = 10^\circ$ between the satellite and ground stations is maintained throughout the simulations. The receiver antennas are characterized by an efficiency of $\eta_{\text{rec}} = 0.6$ and a maximum gain of $G_{\text{rec}} = 35$ dBi. For the state representation in the DQN model, the number of closest visible satellites, V , is set to 10. The simulation parameters are summarized in Table 2.

The composite receive gain used in equation (7) is calculated based on the ITU Recommendation S.1528, which is specifically designed for LEO satellites [32]. Figure 9 illustrates the obtained composite receive gain, with the red curve representing the gain profile for LEO satellites and the blue curve representing the gain profile for GEO satellites. The latter is compared with the gain profile derived from ITU Recommendation S.465-6, which is intended for GEO satellites [33]. As anticipated, the LEO satellite system necessitates ground station antennas with a broader radiation pattern to accommodate the more dynamic link variations inherent to LEO operations.

TABLE 2. Simulation parameters.

Notation	Parameter	Value
h_{sat}	Altitude of Satellites	600 km
f_{sat}	Operating Frequency	19.5 GHz
N_{ant}	Satellite Antenna Type	50×50 planar array
$EIRP_{\text{sat}}$	EIRP Spectral Density	-48 dBW/Hz
B_{GW}	GW Link Bandwidth	2 GHz
B_{UT}	UT Link Bandwidth	200 MHz
D_{GW}	GW Antenna Diameter	3.5 meters
D_{UT}	UT Antenna Diameter	0.6 meters
θ_{min}	Minimum Elevation Angle	10 degrees
η_{rec}	Receiver Antenna Efficiency	0.6
G_{rec}	Receiver Antenna Maximum Gain	35 dBi
V	Closest Visible Satellites	10

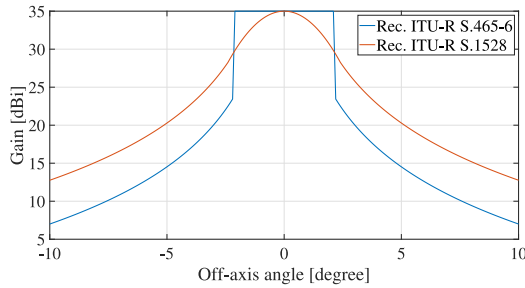


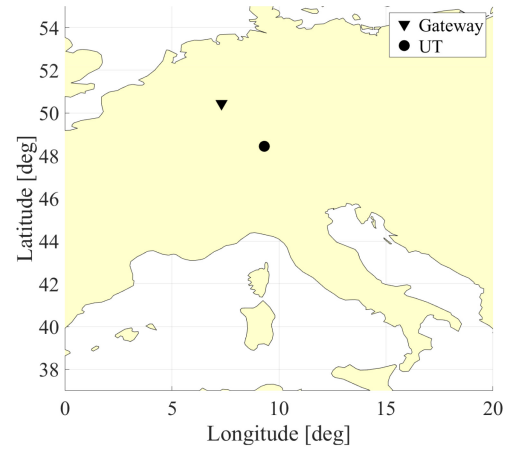
FIGURE 9. Composite receive gain as a function of the off-axis angle.

We consider two simulation scenarios. In the first scenario, we consider one GW and a single target UT, as illustrated in Figure 10(a). For the second, we consider four GWs, a single target UT, and three additional users. The location of the GWs and UTs is presented in Figure 10(b).

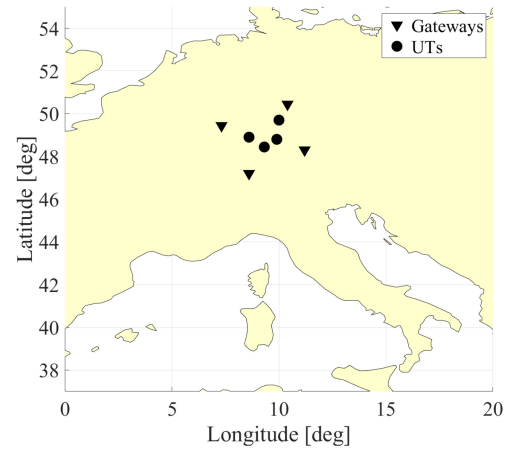
A. SCENARIO 1: ONE GW AND ONE UT

In the first scenario, we focus on a simplified setup involving a single GW and a single target UT. This scenario, depicted in Figure 10(a), allows us to closely analyze the handover dynamics and signal quality in a controlled environment with minimal interference and competition for resources.

Figure 11 presents the evolution of the C/I metric over time for a specified C/I threshold of $C/I_{\text{thresh}} = 20$ dB. In this scenario, all handover methods under consideration successfully maintain the prescribed C/I threshold of 20 dB, ensuring a minimum quality of service. The proposed DQN-based handover strategy outperforms the visibility-based handover approach in terms of achieved C/I, which is expected since the reward function explicitly incorporates a term that rewards improved C/I. Moreover, the DQN-based handover performs close to the optimal performance bound, represented by the best C/I-based handover, which always selects the satellite with the highest C/I ratio. The best C/I-based handover method is impractical because it requires estimating the C/I values for all potential satellites before making a handover decision. However, accurately predicting these values in real-time is challenging due to interference variability and measurement uncertainties. In contrast, the



(a) Scenario 1



(b) Scenario 2

FIGURE 10. GW and UT locations for considered scenarios.

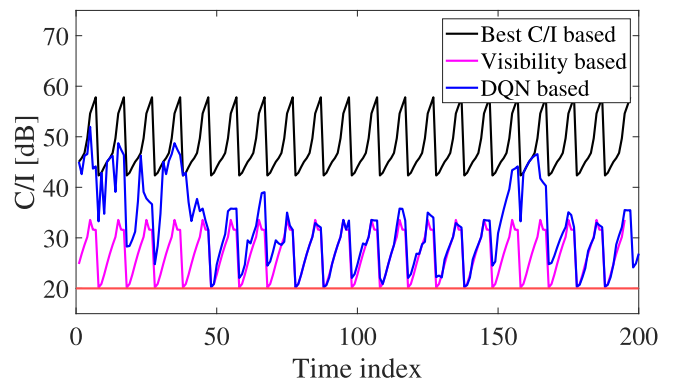


FIGURE 11. Instantaneous C/I (dB) metric for scenario 1, $C/I_{\text{thresh}} = 20$ dB.

DQN-based strategy optimizes handover decisions using observed states, relying on actual C/I values obtained after handovers to refine its policy over time, rather than requiring complete C/I data upfront.

To further illustrate the behavior of the handover strategies, Figure 12 shows the C/I values of each visible satellite

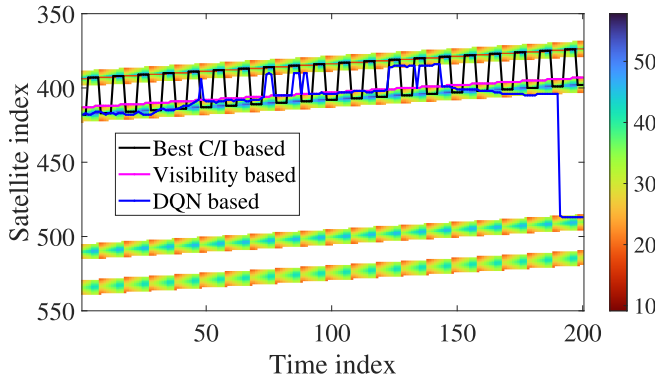


FIGURE 12. Satellite selection for each hand-over method in scenario 1, $C/I_{\text{tresh}} = 20$ dB.

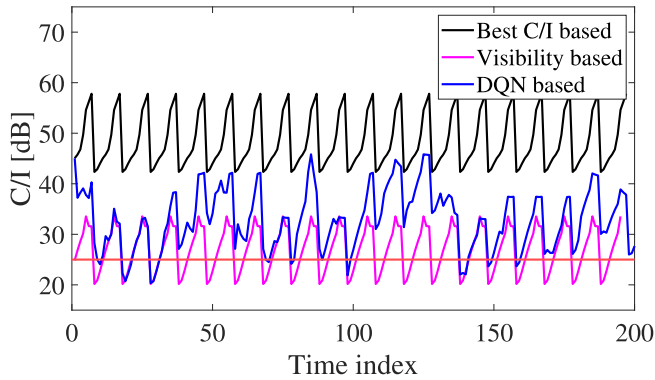


FIGURE 13. Instantaneous C/I (dB) metric for scenario 1, $C/I_{\text{tresh}} = 25$ dB.

at discrete time intervals and how each handover strategy responds. The Best C/I benchmark aggressively switches to whichever satellite currently offers the highest C/I value, which can lead to performance instability due to excessive handovers. In contrast, the visibility-based method remains fixed to a single satellite until its visibility expires, ignoring better C/I alternatives. Our proposed DQN-based method adapts its selection policy over time. Initially, the DQN exhibits exploratory switching, but as learning progresses, it consistently selects satellites that strike a balance between strong C/I performance and longer visibility, resulting in fewer but smarter handovers. This convergence behavior is reflected in the reduction of satellite switching activity at later time indices, indicating that the model has stabilized and is confident in its selections. One noticeable behavior in the DQN satellite selection is a sudden shift in the chosen satellite index around iteration 180. This change reflects a transition point in the learning process, where the agent, having explored various satellite options, converges toward selecting a satellite that offers better long-term C/I performance. Early in training, the model samples different actions to gather experience across multiple states. However, once the DQN estimates become more accurate, it begins favoring satellites with higher predicted cumulative rewards. The sharp drop in the satellite index selected by the DQN-based strategy can be interpreted as the result of a policy

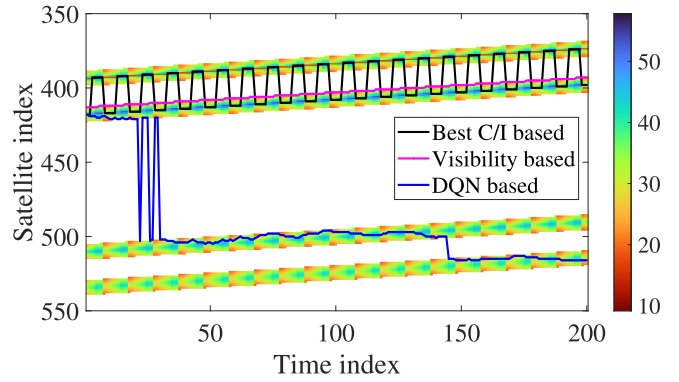


FIGURE 14. Satellite selection for each hand-over method in scenario 1, $C/I_{\text{tresh}} = 25$ dB.

update that balances remaining visibility and signal quality, as reflected in the reward function design. The model learns over time to anticipate future degradation in link quality and proactively switches to a satellite with a more favorable long-term profile.

We then increased the threshold C/I_{tresh} to 25 dB, creating a more challenging scenario for handover management. The results, shown in Figure 13 and Figure 14, reveal that the proposed DQN-based handover model significantly outperforms the visibility-based handover method under this stricter requirement. The DQN approach consistently maintains a higher C/I ratio, demonstrating its robustness in scenarios where signal quality demands are elevated. However, it is worth noting that the increased threshold also introduced a longer convergence period for the RL algorithm, requiring approximately 150 iterations to stabilize. This indicates that while the DQN model adapts effectively to more stringent conditions, there is a trade-off in terms of the time needed for the algorithm to fully converge and optimize the handover decisions. This suggests that while the DQN model adapts effectively to more stringent conditions, there is a trade-off in terms of the time needed for the algorithm to fully converge and optimize the handover decisions. We highlight that temporary drops in the DQN performance curve are expected due to the learning-based nature of our approach. Unlike the Best C/I method, which has full knowledge of all C/I values, the DQN agent relies on trial-and-error learning without access to ground truth before decisions. These fluctuations, particularly during exploration, are part of the RL process and reflect the trade-off between short-term performance and long-term policy optimization. Overall, the agent converges to a near-optimal handover strategy over time.

In contrast, the C/I method, while relatively quick in making decisions once a threshold is crossed, has the major limitation of requiring constant and accurate estimation of the C/I ratio. This ongoing monitoring is challenging and can be error-prone, especially in dynamic environments where interference levels fluctuate. Additionally, the method's reliance on threshold comparison can result

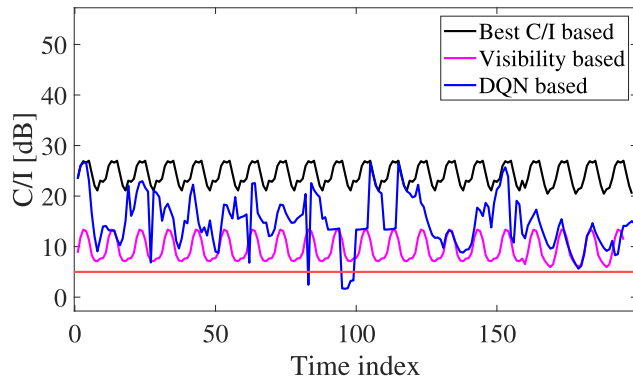


FIGURE 15. Instantaneous C/I (dB) metric for scenario 2, $C/I_{\text{tresh}} = 5$ dB.

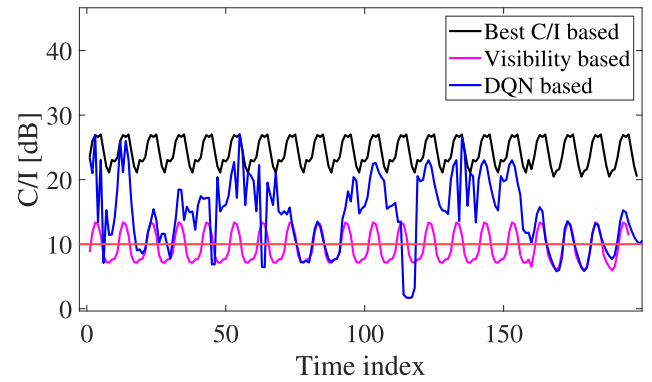


FIGURE 17. Instantaneous C/I (dB) metric for scenario 2, $C/I_{\text{tresh}} = 10$ dB.

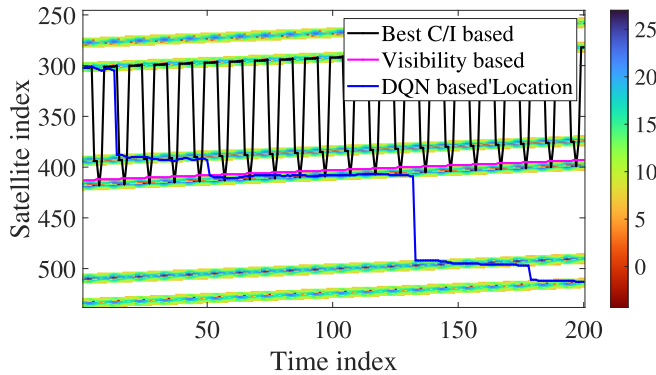


FIGURE 16. Satellite selection for each hand-over method in scenario 2, $C/I_{\text{tresh}} = 5$ dB.

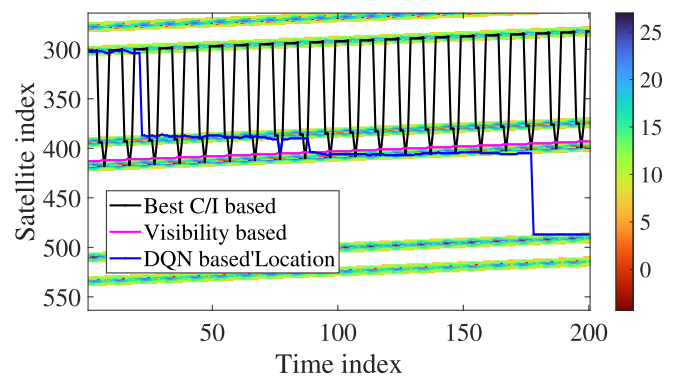


FIGURE 18. Satellite selection for each hand-over method in scenario 2, $C/I_{\text{tresh}} = 10$ dB.

in delayed responses to deteriorating link conditions, leading to unnecessary handovers once the threshold is violated, which can increase signaling load and cause instability in the communication link. This highlights the C/I method's difficulty in providing efficient and timely handover decisions, especially under strict signal quality requirements.

B. SCENARIO 2: FOUR GWS AND FOUR UTs

The second scenario introduces a more complex and realistic network environment. It involves four GWs, a single target UT, and three additional UTs operating simultaneously. The geographical distribution of these GWs and UTs is shown in Figure 10(b). This setup simulates a more interference-limited scenario, where multiple GWs are closely located to the UTs and competing for the same spectral resources, leading to increased potential for interference and a higher frequency of handovers.

Figure 15 presents the evolution of the C/I metric over time for a specified C/I threshold of $C/I_{\text{tresh}} = 5$ dB for the second scenario. After 100 iterations, the proposed DQN-based handover strategy demonstrates superior performance compared to the visibility-based handover approach. The number of iterations required for convergence can be attributed to the exploration-exploitation trade-off in reinforcement learning, where the model initially explores a wide

range of actions before optimizing its policy. Additionally, the complexity of the interference scenario and the delayed reward feedback contribute to the need for more iterations to achieve stable and optimal performance. Moreover, the DQN-based handover performs close to the optimal performance bound, represented by the best C/I-based handover, which always selects the satellite with the highest C/I ratio. In Figure 16, we can see the satellites been selected by each handover strategy as time pass.

We then increased the threshold C/I_{tresh} to 10 dB, creating a more challenging scenario for handover management. The results, shown in Figure 17 and Figure 18, reveal that the proposed DQN-based handover model significantly outperforms the visibility-based handover method under this stricter requirement. The DQN approach consistently maintains a higher C/I ratio, demonstrating its robustness in scenarios where signal quality demands are elevated. However, it's worth noting that the increased threshold also introduced a longer convergence period for the RL algorithm, requiring approximately 150 iterations to stabilize. This indicates that while the DQN model adapts effectively to more stringent conditions, there is a trade-off in terms of the time needed for the algorithm to fully converge and optimize the handover decisions.

The proposed DQN-based handover strategy has been evaluated against classical heuristic methods, namely the

C/I-based and visibility-based handovers, which serve as widely accepted baselines in the literature. Although state-of-the-art RL-based approaches such as [26], [27] offer valuable performance benchmarks, their applicability to our scenario is limited. These methods typically address different problem formulations, such as generic connectivity optimization or centralized multi-user access control, and do not consider interference mitigation under spectral coexistence between feeder and user links. Furthermore, they often assume centralized architectures or access to global network information, whereas our work focuses on distributed decision-making under partial observability. Incorporating a fair comparison under unified assumptions will be explored in future work.

While the presented simulations focus on scenarios with one and four UTs/GWs, these configurations were chosen to enable clear interpretation of the handover decisions and their interference-driven dynamics. They offer valuable insight into the behavior of the learning model in both low- and moderate-interference environments. Nonetheless, we acknowledge that evaluating the system at a much larger scale is essential for deployment in operational LEO constellations. Our DQN-based handover approach is inherently distributed, requiring only local observations, which makes it suitable for scaling to hundreds or thousands of UTs. Future work will extend the simulation environment to include large-scale satellite constellations and higher UT densities, allowing us to study convergence time, system performance, and interference coordination under realistic traffic loads.

V. CONCLUSION

In this work, we proposed a Deep Q-Network (DQN)-based handover strategy for LEO satellite communications. The aim was to reduce interference between gateway and user terminals by using machine learning techniques to dynamically select the optimal satellite for user terminals during the handover process. Our results suggest that the DQN-based approach can adapt to the dynamic nature of LEO satellite constellations, making it a robust solution for future satellite communication networks.

Future work could explore the integration of this approach with other network optimization techniques and assess its performance in even more complex and variable operational environments.

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Please note that the views of the authors of this paper do not necessarily reflect the views of SES S.A.

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